

Xiyu Gu

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Education

Jan.2024 – Present

University of Padua

Padua, Italy

Ph.D. Candidate

Research Area: Computation–Communication–Control Co-design

Supervisor Prof. Matthias Pezzutto

Co-supervisor Prof. Luca Schenato

June2022 – Dec.2023.

University of Victoria

Victoria, Canada

Research Assistant

Supervisor Prof. Yang Shi

Sep.2019 – Mar.2022

Northwestern Polytechnical University

Project 985 & 211, Xi'an, China

Master of Engineering

Major: Navigation, Guidance & Control

GPA: 3.55/4.00, rank 3/69

Supervisor Prof. Jianguo Guo

Co-supervisor Prof. Zongyi Guo

Sep.2015 – Jun.2019

Northwestern Polytechnical University

Project 985 & 211, Xi'an, China

Bachelor of Engineering

Major: Flight Vehicle Design & Engineering

GPA: 3.58/4.00, rank 2/32

Research Interests

- Performance-oriented optimal control of networked systems
- Prescribed performance and safety-constrained control
- Applications to flight vehicles and transportation systems

Publications

Journal Papers.....

- [1] X. Gu, L. Schenato, S. Dey, M. Pezzutto, Communication-Computation Trade-off and Optimal Control Selection in LQG Control, *IEEE Control Systems Letters*, 2025.
- [2] X. Gu, Z. Guo, X. Wang, et al, Prescribed Performance-Guaranteed Tracking Control: A Switched Positive System Perspective. *IEEE Control Systems Letters*, 2024.
- [3] Z. Guo, X. Gu, Y. Han, et al, Performance-Guaranteed Attitude Tracking Control for RLV: A Finite-Time PPC Approach. *IEEE Transactions on Aerospace and Electronic Systems*, 2024. (**Corresponding author**)
- [4] Z. Guo, D. Herry, X. Gu, et al, Maneuvering tracking algorithm for reentry vehicles with guaranteed prescribed performance, *IEEE Transactions on Aerospace and Electronic Systems*, 2024, 60.3: 3012-3020.
- [5] X. Gu, J. Guo, Z. Guo, Performance improvement-oriented reentry attitude control for reusable launch vehicles with overload constraint, *ISA Transactions*, 2022, 128: 386-396.
- [6] J. Guo, X. Gu, Z. Guo, Asymptotic adaptive tracking control for hypersonic vehicles with guaranteeing multi-performance requirements. *Aerospace Science and Technology*, 2020, 105: 106025.

Conference Papers.....

[1] X. Gu, S. Dey, M. Pezzutto, Cloud-based Reference Governor with Dynamic Computation Offloading over Shared Computing Resources, *European Control Conference (ECC)*, 2026.

[2] X. Gu, M. Pezzutto, L. Schenato, S. Dey, Optimal Control Selection over the Edge-Cloud Continuum, *IFAC- PapersOnline*, 2025. (**Finalist for Young Author Award**)

Under Review.....

[1] Y. Dong, Z. Guo, Z. Peng, X. Gu, Data-Physics Fusion-Guided Bayesian Fuzzy Learning for Interpretable Reentry Vehicle Trajectory Prediction, *IEEE Transactions on Aerospace and Electronic Systems*, under review, 2026. (**Corresponding author**).

[2] X. Gu, M. Pezzutto, L. Schenato, S. Dey, Cooperative Edge-Cloud MPC for Constrained Control under Arbitrary Communication Environment, *Automatica (regular paper)*, under review, 2025.